THE SET COLIMIT OF A DIAGRAM OF ABELIAN GROUPS SURJECTS ONTO THE ABELIAN COLIMIT

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Editorial note. Counted span is the author's own proof environment, source lines 2523-2535, following the statement at 2512-2521.

Editorial note. The pair case is executed in the span; arbitrary finite sums reduce to it by induction on the same move, and the quotient-relation compatibility is part of the recalled constructions, both recorded in the card's obligations.

Allowed standard prerequisites (editorial).

- Colimits in Set as quotients of the disjoint union. section-limit-sets, cited at proof lines 2524-2526. Supplies the concrete description of the set colimit used as the source of the surjection.
- Colimits in Ab as quotients of the direct sum. the recalled description at proof lines 2527-2528. Supplies the concrete description of the abelian colimit whose elements are represented by finite sums.

Source reference. Original section 002U: sources/stacks-categories/pinned-original-source.tex, source lines 1807–1807. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{I}$ be an index category, i.e., a category. Assume that for every pair of objects x, y of $\mathcal{I}$ there exist an object z and morphisms $x \to z$ and $y \to z$. Let $M : \mathcal{I} \to Ab$ be a diagram of abelian groups over $\mathcal{I}$. Then the colimit of M in the category of sets surjects onto the colimit of M in the category of abelian groups.*

Proof. Recall that the colimit in the category of sets is the quotient of the disjoint union $\coprod M_i$ by relation, see Section 002U. Similarly, the colimit in the category of abelian groups is a quotient of the direct sum $\bigoplus M_i$. The assumption of the lemma means that given $i, j \in \mathrm{Ob}(\mathcal{I})$ and $m \in M_i$ and $n \in M_j$, then we can find an object k and morphisms $a : i \to k$ and $b : j \to k$. Thus $m + n$ is represented in the colimit by the element $M(a)(m) + M(b)(n)$ of M_k. Thus the $\coprod M_i$ surjects onto the colimit. □

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